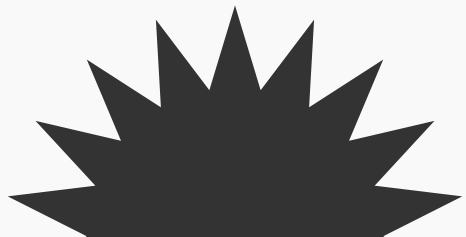


Bias & AI



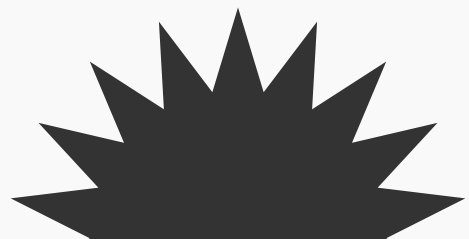
What is Bias?

Bias in AI occurs when an AI system produces **systematically** unfair outcomes due to **prejudiced** assumptions embedded in the data or design.



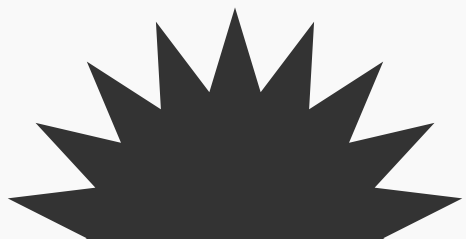
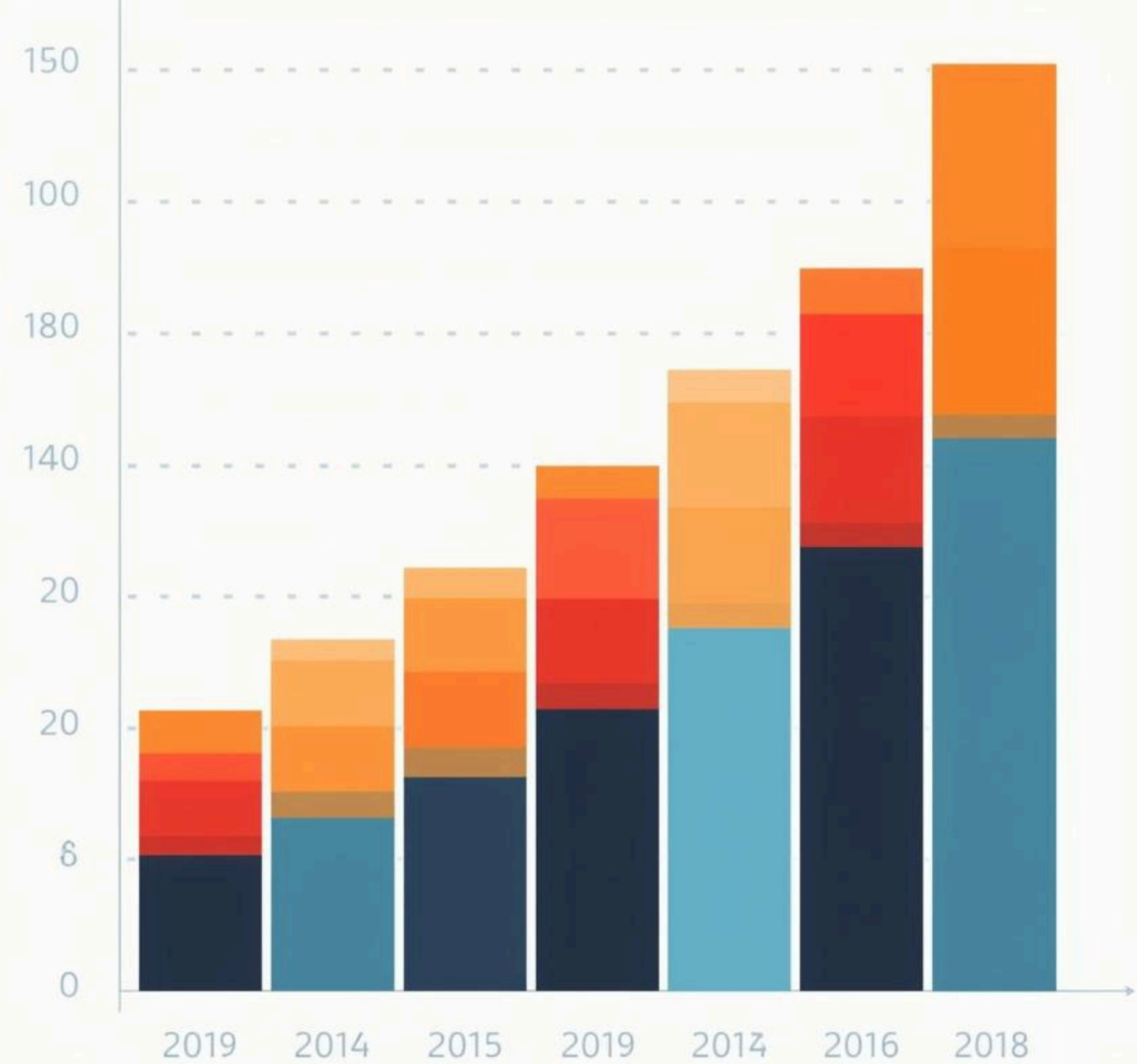
Types of Bias?

- Data
- Historical
- Measurement
- Algorithmic
- Deployment
- Interaction
- Confirmation



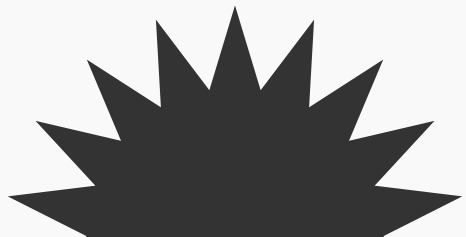
Understanding Data Bias in AI

- Training data is unbalanced or incomplete
- Underrepresentation or overrepresentation of groups
- “Garbage in, garbage out” at scale



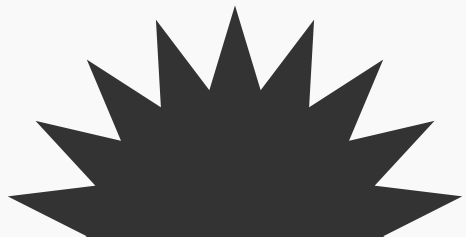
Understanding Historical Bias in AI

- Even “correct” data reflects unjust social conditions
- Past discrimination becomes future prediction
- AI trained on history will reproduce history



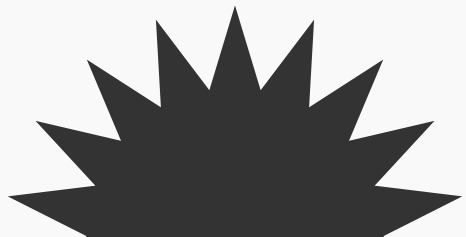
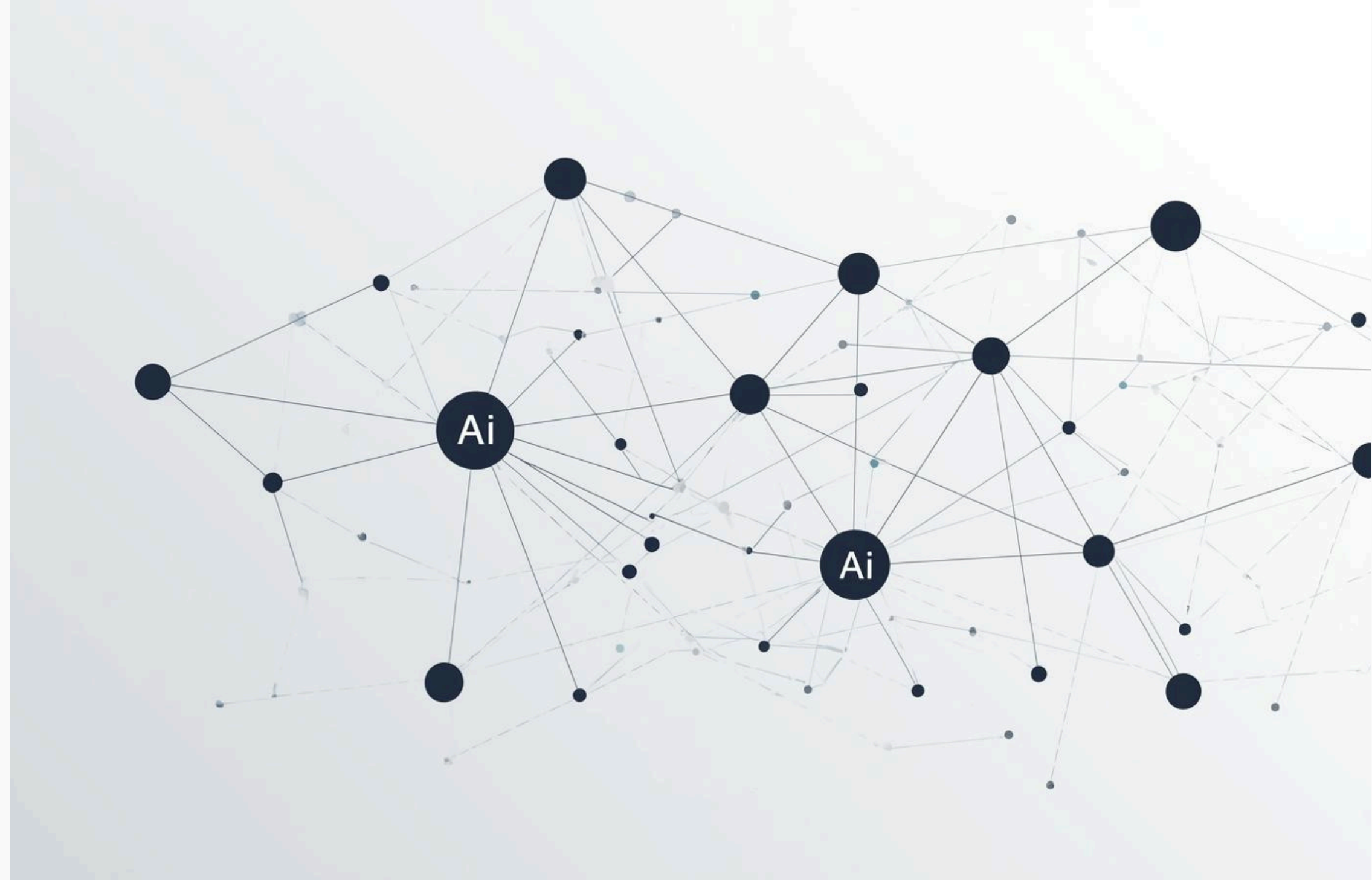
Understanding Measurement Bias in AI Systems

- Proxies used instead of direct measurements
- Example: ZIP code used as a proxy for creditworthiness



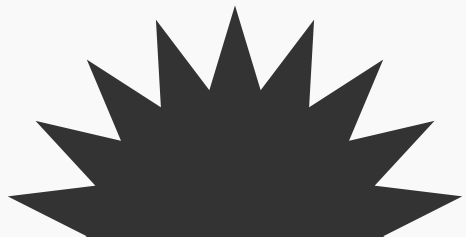
Understanding Algorithmic Bias in AI

- Model design choices amplify disparities
- Optimization goals unintentionally favor one group



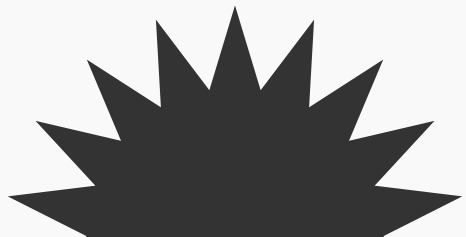
Understanding Deployment Bias in AI

- AI systems are used outside their intended context
- Human behavior changes once the system is in place



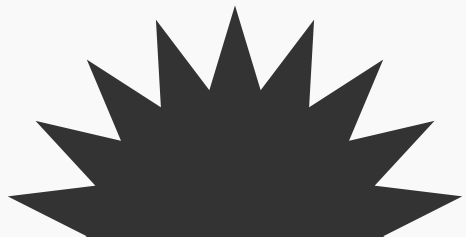
Understanding Interaction Bias in AI

- Bias emerges through ongoing interaction between users and AI
- The system learns from user behavior, prompts, or feedback
- Harmful patterns are amplified through repeated use



Understanding Confirmation Bias in AI

- AI systems reinforce existing beliefs, patterns, or assumptions
- Models are trained or tuned to confirm what decision-makers already expect
- Feedback loops reward predictions that align with prior judgments



Impact of Bias

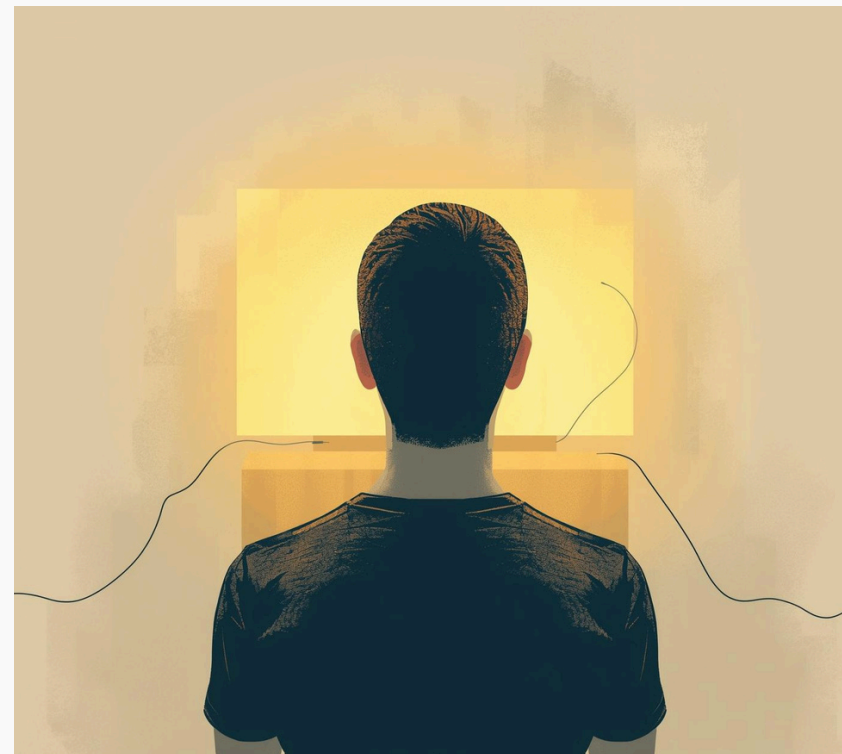
Discrimination

Bias can lead to unfair treatment of individuals.



Trust Loss

People may lose faith in AI systems overall.



Legal Challenges

Biased AI can lead to serious legal issues.

